

Week 2 — Stability & Intelligence

Goal: Add tracking, temporal stability, and intelligent triggering.

Day	Key Tasks
Day 1	Understand ByteTrack, Track IDs across frames
Day 2	Temporal EMA smoothing, Reduce flickering detections
Day 3	SSIM scene change detection, Adaptive narration trigger
Day 4	Scene buffer architecture, Shared state management
Day 5	Threading basics, Inference/display separation
Day 6	Performance profiling, Latency optimization
Day 7	Stress testing, Static vs dynamic scene evaluation

Expected Deliverables

- Stable tracked objects
- Reduced narration flicker
- Adaptive VLM triggering
- Threaded pipeline

This week contributes directly toward building a complete real-time scene understanding research system capable of object detection, depth estimation, segmentation, narration, and intelligent perception.